

APM Project Management Qualification

Learner Study Pack



Schedule management

Introduction

This section of the Learner Study Pack will cover the Schedule Management competence of the APM Project Management Qualification.

This section will enable you to gain an understanding of what schedule management is, defining scope and relationships between activities for estimating, and the benefits of re-estimating throughout the project life cycle.

Learning objective and outcomes

Learning objective:

20. Understand schedule management as the ability to undertake time-based planning with an emphasis on activities and resource.

By the end of this activity, you should be able to:

20a) Know how to define scope in terms of outputs, outcomes, and benefits (including use of product, work and cost breakdown structures).

20b) Understand there are links and dependencies between activities within a project and business-as-usual activities, for example business-as-usual activities, costs, quality, risks, and scope can all impact the schedule for the project.

20c) Understand the reasons for and benefits of re-estimating, and schedule optimisation, throughout the project life cycle.

Study time

This section of the Learner Study Pack will take approximately 1 – 2 hours to complete.

Recommended resources

In addition to your Learner Study Pack, use the below references for more information on Schedule Management:

- APM Body of Knowledge Seventh edition, 4.2.5 (Scheduling – critical path); 4.2.6 (Scheduling – critical chain)
- APM Interactive Learning Module, Project Controls: Scheduling, Accessed at: APM Learning
- APM Interactive Learning Module, Planning, Scheduling, Monitoring and Control – The practical project management of time, cost, and risk, Accessed at: APM Learning
- APM Resources, What is scheduling in project management? Accessed at: <https://www.apm.org.uk/resources/what-is-project-management/what-is-scheduling-in-project-management/>
- APM News, Schedule Integrity – The key to successful project management. Accessed at: <https://www.apm.org.uk/news/schedule-integrity-the-key-to-successful-project-management/>

Overview

Schedule management is a fundamental part of project management which focused on the effective and efficient allocation of time and resources.

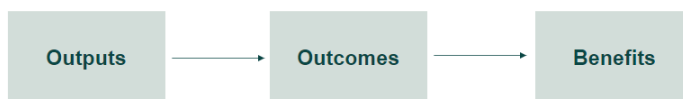
Schedule management is:

- Used to develop schedules of when work is planned to be completed in projects.
- Time-based planning with an emphasis on activities and resource.
- The process for developing and maintaining schedules.
- When work for a particular project is planned to be done.
- Consideration of dependencies and can be for internal and / or external resources and activities.

Defining Scope

Scope is defined by the outputs, outcomes, and benefits being delivered and a high-level scope should normally be recorded in the business case.

A project's outputs are the tangible or intangible deliverables being created which produce the outcomes of changed circumstances, behaviours or practices, to create the benefits in line with the defined acceptance criteria.



Breakdown Structures

These three models can be used to illustrate what needs to be done to ensure that the projects outputs are meet:

- Product Breakdown Structure (PBS)

A PBS identifies the scope of a project to be delivered

The end output of the project is placed at the top level, defining all components as you move down levels until reaching the individual products to be delivered

Each product has defined acceptance criteria and quality control methods

This can then be used to secure stakeholder agreement of the deliverables and agree what is in and out of scope

- Work Breakdown Structure (WBS)

A WBS defines a baseline scope of work capturing the activities or work to be completed to deliver the project requirements

The lowest level of detail identified is a work package

Looking at the work required, and skill set needed to complete

Work requiring a similar skill set can then be grouped this supports allocation of resources, estimating and scheduling of the work packages

- Cost Breakdown Structure (CBS)

A CBS provides a financial view by breaking down the project into cost elements

It can be used to show either the costs assigned to work packages, using the WBS such as people, equipment, materials or other required resource, or the costs assigned to functional areas and third parties using the organisational breakdown structure (OBS)

This can then be used to track project spend to understand achievement against the estimates produced

Links and Dependencies

Part of effective schedule management is understanding the impact of links and dependencies on project tasks, decision gates and other milestones.

Links represent the relationships between different activities outlining the order in which they should occur, whereas dependencies define the reliance of one task on another for successful completion.

Effective schedule management should also consider the various links and dependencies between activities within a project and business as usual activities.

These may include:

- Relationships between events – the links between activities and their implications
- Stakeholders – from a technical and BAU perspective to inform on potential impacts to time / resources
- Scope definition – prevent risk of delays through misunderstanding of what is in and out of scope for the project
- Business change – activities that are resourced by BAU, not the project
- Dependencies – between different project activities and BAU
- Schedules – advise of activities and any potential conflicts requiring resolution

Re-estimating and Schedule Optimisation

An estimate is just that an estimate - an approximation of time to complete tasks.

The activity undertaken to complete estimating tasks should not be a one off but instead ongoing as more information is known.

Re-estimating will increase confidence in the estimates being used to create the schedule.

As part of the project life cycle, the following will ensure schedule optimisation:

	Actions	Benefit
Greater involvement of the project team	Apply technical expertise to clarify what is needed to achieve project scope. Minimise the effect of any previous or ongoing estimating errors.	Increase the accuracy of the estimates through technical expertise. Providing increased confidence in the project's ability to deliver against the success criteria.
Opportunity to incorporate lesson learnt and reduce contingency reserves	Validate assumptions as scope is clarified to reduce uncertainty. Include variances and trends from previous work to remaining estimates to adjust for ongoing activity to incorporate findings.	Remove reserves that are no longer needed to other projects, as forecasts become more realistic to actual circumstances.
Realisation of an accepted risk	Understand the impact of the realised risk on the project.	Increase awareness of the situation and any potential impact with key stakeholders and, if necessary, request to draw down contingency for the project to continue.
Increased likelihood of adhering to overall estimate	Provide visibility of variances to be able to inform stakeholders and take any necessary actions. Apply productivity data for better understanding of final costs and timescales.	Increase confidence in the project's performance and provide information that supports any required escalations for further funding or time for the project to deliver against the success criteria.

Understanding the concepts that feed into the activity are key - WBS, estimating, dependencies, resource allocation, alignment of success criteria, and baselining are the backbone to scheduling and optimisation of that schedule for a successful project.

Critical Path and Critical Chain

There are two primary methods used to support scheduling techniques:

1. Critical path

The critical path method places emphasis on the shortest time to complete all activities in a logical order. The logic can include whether activities can start at the same time, if one activity needs to be complete before the next begins etc. Identifying the relationship or inter-dependency between tasks is essential in identifying the critical path.

2. Critical chain

The critical chain method places emphasis on the resources attempting to keep them at a constant utilisation

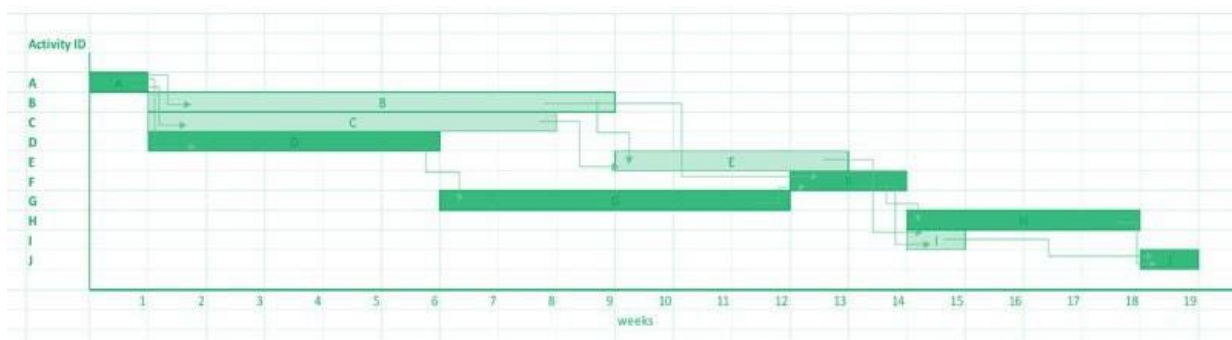
If resources were unlimited and always available, the critical path and critical chain method for scheduling would provide the same result.

The results of these methods are usually presented as a visualisation e.g. a Gantt chart that shows either activities or resources as bars on a timeline.

Critical Path

- Identifies the activities, or tasks in the plan that if delayed will affect the project end date.
- These tasks create the longest pathway of activity on the project and therefore become the critical activities to be managed.
- To identify the critical path, you also need to understand the interdependencies between activities, the duration of time each activity is estimated for and whether any float/slack has been planned in.
- The start and finish of activities are the focus for critical path reporting.

Below is an example of a Critical Path visualisation:

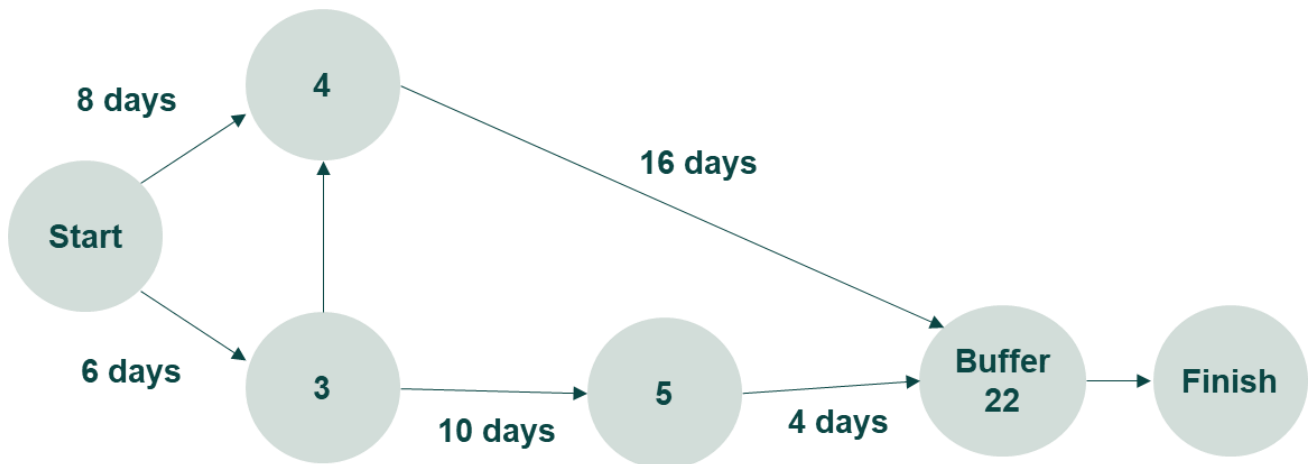


Critical chain

- Various factors such as, skills, abilities, tools and motivation influence work rate, the resulting durations and the effects on the next planned activities.
- The critical chain method takes the estimated durations for each activity and reduces it by half to create a time contingency.

- This is added up for all the activities to create a buffer at the end for the chain of activities rather than the individual activities.
- This is done to balance out the effect of tasks finishing early or later than planned to prevent losing the benefit of tasks completing early or the impact if late to the project.
- The buffer is there to protect the whole of the project.

Below is an example of a Critical Chain visualisation:



Scenario One:

The renovation project is set to get underway in less than four weeks and Robin, the Finance Manager is started to get concerned about the incredibly short construction schedule that the main contractor has presented to them.

As the member of the senior leadership team that has been asked to oversee the allocation of funds and the overall budget, they are worried that the contractor hasn't followed schedule management techniques properly and that some of the timings for the work aren't accurate and that steps may have been missed altogether.

What elements should be considered to make a schedule effective?

Scenario Two:

You've reviewed the schedule that the main contractor has submitted to Robin, and it appears quite basic in its structure and considerations of the scope and the desired outcomes for the renovation project.

To ensure that construction begins without any delay, you decide to add more detail to the schedule to ensure that everyone is clear on what is required and when it needs to be delivered.

This will be a large piece of work and there isn't long until the project gets underway, but this will ensure the project is far more likely to run smoothly and will be delivered in the agreed timeframes.

How can each of the three breakdown structures support effective scheduling and why?

What are some examples of the links and dependencies that should be considered as part of the schedule?

Scenario Three:

The original cost estimates for the renovation project were based on assumptions about material prices, labour costs and the regulatory requirements of working on an historic building.

However, as the first day of construction gets closer and closer, there are reports from the main contractor that the soil conditions around the building may require additional foundation work and the project may have to be re-estimated.

What could be the positive impacts of re-estimating the project at such an early stage?

What are the potential repercussions if the project is not re-estimated at this stage?

Learning review questions

Before you attempt the exam practice questions in the next section, we recommend you review your learning to quickly see which aspects you don't understand or can't remember as well.

It will also boost your confidence by confirming what you have learnt and your demonstrated ability to apply that knowledge, so that you are ready to practise for the exam.

Here are some questions you may wish to answer to review your learning from the Leadership section:

- What are the three different Breakdown Structures and how do they support effective scheduling?
- How could a Work Breakdown Structure (WBS) support the creation of a schedule?
- How can re-estimating either positively and/or negatively impact the schedule for a project?
- What are the benefits of understanding the links between project and BAU activities for the schedule?

Exam practice questions

20.1 Which combination of the below would assist in defining the scope of a project?
(1 mark)

1. Product Breakdown Structure
2. Project Breakdown Structure
3. Work Breakdown Structure
4. Resource Breakdown Structure
5. Cost Breakdown Structure
6. Communication Breakdown Structure

Select the correct combination of three options:

- a) 1,3,5
- b) 2,3,6
- c) 2,4,5
- d) 1,4,6

20.2 You are the project manager for creating a prototype healthcare device which is required for a pioneering surgical procedure. The timeframe for the project is fixed because the device is required by the client on a specific date.

During the project definition stage, you have identified a critical dependency related to obtaining regulatory approval of the device. This is required prior to being able to handover to business-as-usual. Any delay in achieving approval has the potential to impact the overall project schedule and there's also a high risk of changes in scope due to evolving regulatory requirements.

What strategic approaches should you adopt to manage this situation?
(1 mark)

1. Fast-track other project activities to compensate for potential delays.
2. Engage with regulatory authorities to expedite the approval process.
3. Develop contingency plans for possible scope changes and schedule adjustments.
4. Communicate the potential risks and delays to stakeholders proactively.
5. Identify additional resources that could support timely completion.
6. Capture the need for approval as an outstanding action in the transition plan

Select the correct combination of statements:

- a. 1, 2, 4
- b. 3, 4, 5
- c. 2, 4, 6
- d. 1, 3, 5

20.3 Why is it often important to re-estimate the project schedule at the end of the Concept phase?
(1 mark)

- a. The project sponsor will expect commitment to a project end date at the Concept phase decision gate.
- b. The project team is now formed and has started more detailed planning, meaning that estimates can be more accurate.
- c. The end users will have reviewed the potential solutions and updated their requirements, meaning that the scope can be more accurate.
- d. Responses to risks will have been actioned, meaning that the estimates can be more accurate.

20.4 What are two reasons why it is important to clearly define the project scope at an early stage of a project?
(2 marks)

20.5 What are five ways that activities in business-as-usual (BAU) can impact a project's schedule?
(5 marks)

20.6 You are managing a project to build a new block of apartments in a large city. You are in the design phase of the project and have completed a PESTLE analysis of your project environment. From this analysis, you identify potential supply chain disruption as a key event within your external environment.

Based on the information in the question, describe two reasons you may decide to conduct regular re-estimations of the project's schedule and costs.
(2 marks)

Describe three benefits the project team will gain by completing these regular re-estimations.
(3 marks)

Exam practice answers

Question	Answer / mark scheme
20.1	A
20.2	B
20.3	B
20.4	1 mark for each reason why it is important, 2 marks available in total – any of the below: • Avoids misunderstanding / misalignment on what the project is delivering. • Can reduce amount of future change requests. • Ensures that stakeholder expectations can be met / avoids disappointing / disengaging stakeholders at a later date. • Can reduce risk of project being delayed./enables effective planning • Can reduce risk of project going over budget.
20.5	1 mark each for any of the below, 5 marks available in total. For example, but not limited to: • Equipment can be required for BAU and not be available for the project, leading to schedule delays. • Dependencies on a BAU output that is delayed can delay a project schedule in turn. • Conflicts between the needs of BAU and the needs of a project can impact a project schedule. • Stakeholders can be required for BAU and not available when the project needs, delaying a project schedule. • The project team could be required to balance their work on BAU with the project, which means that the schedule may need to include BAU activities as dependencies. • Confusion over the project scope can lead to stakeholders expecting BAU activities to be delivered by the project.
20.6	1 mark to be awarded for each reason why supply chain disruption would lead to re-estimations, including, but not limited to the below (maximum of 2 marks): • Deliveries of materials could be delayed so you could need to extend the schedule. • Materials could be scarcer, and take longer to source, so you could need to extend the schedule. • Materials could be scarcer, and therefore more expensive, so you could need to increase the budget. 1 mark to be awarded for each benefit presented from the list below (maximum of 3 marks) • Reduced contingency reserves needed. • Greater involvement of the project team. • Opportunity to incorporate lessons learnt. • More opportunities to assess and respond to risks. • More opportunities to make decisions and escalate problems. • Increased likelihood of adhering to original estimate. • Minimise the effects of estimating errors. • Greater overall control of the project. • Better capability to respond to changes on the project.